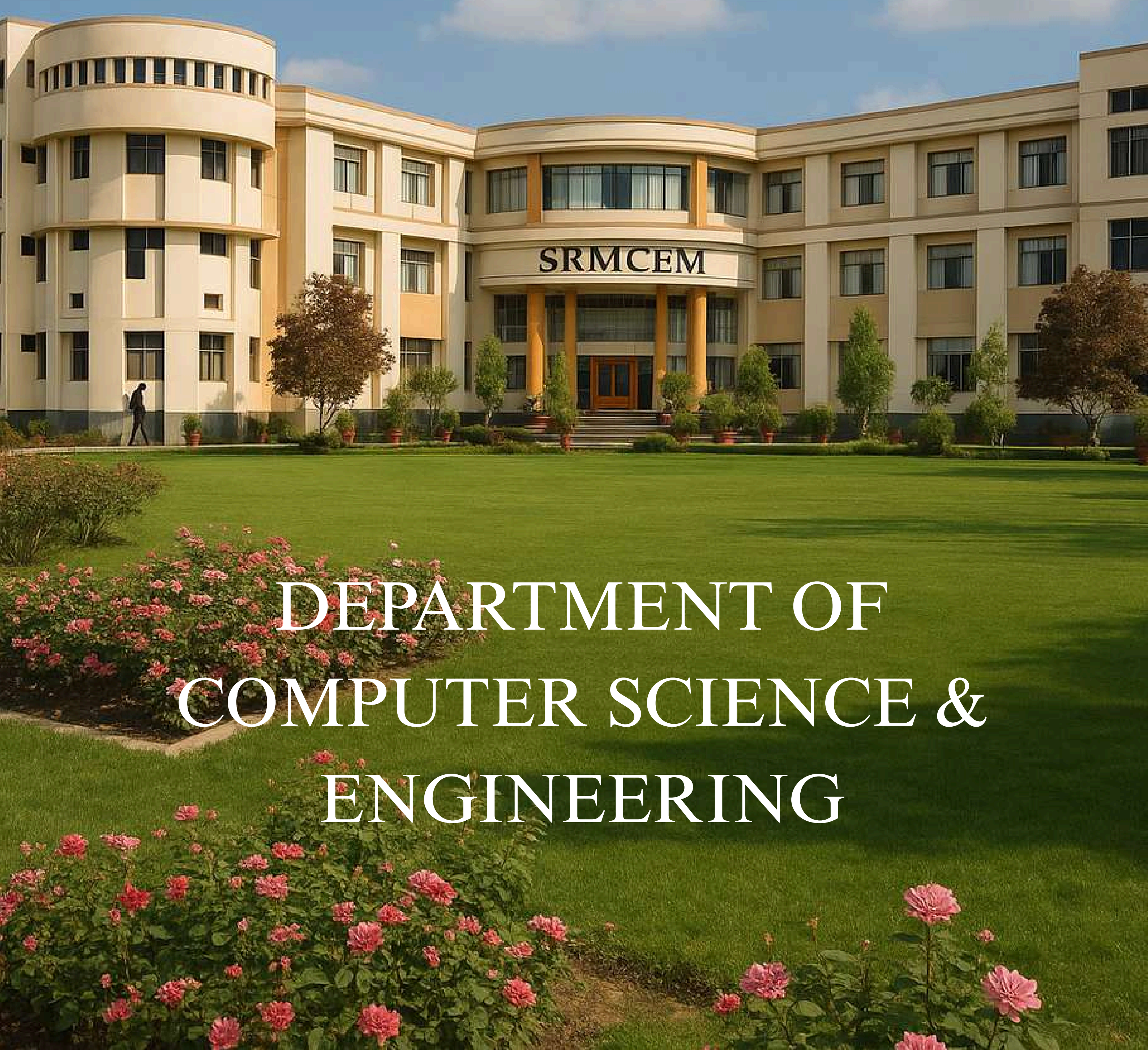
SHRI RAMSWAROOP

MEMORIAL COLLEGE OF ENGINEERING & MANAGEMENT



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

PREFACE

"Simplicity, when taken to its finest form, becomes elegance."

This thought perfectly echoes our belief that every sincere effort, when made with integrity, ultimately leads to the most rewarding outcomes.

As we put together this magazine, we revisited countless memories filled with dedication, collaboration, and the collective hard work of the entire CSE department.

With great pride, we present to you the 6th edition of ABHIPSA: A Glimpse into the CSE Department for the academic year 2023–2024.

This edition serves as a reflection of the vibrant spirit of our department, showcasing the events, achievements, and initiatives we've undertaken throughout the year. It stands as a tribute to the shared efforts and passion of both students and faculty members.

We sincerely thank our readers for their continued encouragement and support. It is the generous sharing of ideas and wisdom that has brought this publication to life.

Our heartfelt gratitude goes to the editorial team for their dedication and hard work in making this magazine a reality.

Thank You

TABLE OF CONTENT

ABOUT

EVENTS

In the memory of

SHRI RAMSWAROOP AGARWAL JI

1925-1980

ER. PANKAJ AGARWAL
EXECUTIVE DIRECTOR

ER. POOJA AGARWAL
ADDITIONAL EXECUTIVE DIRECTOR

ABOUT US

Shri Ramswaroop Memorial College of Engineering & Management, Lucknow is an ISO 9001:2008 certified, AICTE recognized, and Dr. APJ Abdul Kalam Technical University, Lucknow affiliated institution. Set in placid surroundings on the Lucknow-Faizabad Highway, the group of institutions offers the best in engineering and management education.

It provides the opportunity of studying and training under experienced faculty, and is equipped with stateof-the-art classrooms, labs, and library facilities in the quiet and cultural milieu of Lucknow.

Established by two IIT Kanpur gold medallists, Mr. Pankaj Agarwal and Mrs. Pooja Agarwal, the group of institutions is driven by the dream of building a future on the strength of youth fortified with the best professional training and education. The striving for excellence incessantly goes on at SRMCEM. The integrity of its managerial leadership and faculty, value-based institutional culture, and quality-driven performance initiatives make it a promising place to acquire professional education.

COLLEGE VISION

To achieve international standards in value based professional education for the benefit of society and the nation.

COLLEGE MISSION

- To devote teaching, learning, and collaboration to the pursuit of frontier technologies with an innovative and exceptional spirit.
- To foster human values and ethos, compassion for the ecosystem and obligation towards society and the nation.
- To provide an environment conducive to continuous learning and all-round development of the college fraternity

DEPARTMENT VISION

To become a World Class seat of Learning in Computer Science to produce Competent Software Professionals with Strong Values and Dedication to the Nation.

DEPARTMENT MISSION

M1: To Produce Competent Computer Science Professionals through Quality Education.

M2: To Inculcate Social and Ethical Values in students for the Well being of the Nation.

M3: To Encourage Exploration of Cutting Edge Technologies and Pursuance of Lifelong learning


DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING
NATIONAL BOARD OF ACCREDITATION
NBA
ACCREDITED BY
Congratulations!
PROUD MOMENT FOR RAMSWAROOP GROUP
SRMCEM
BECOMES THE ONLY CAMPUS TO GET
NBA
NATIONAL BOARD of ACCREDITATION

PROGRAMME OUTCOMES

Engineering Graduates will be able to

- PO 1. Engineering Knowledge: – Apply the knowledge of mathematics science, engineering fundamentals, and an engineering specialization to a solution of complex engineering problems.
- PO 2. Problem analysis:- Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
- PO 3. Design / Development of solutions:- Design solutions for complex engineering problems and design systems components or processes that meet specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.
- PO 4. Conduct investigation of complex problems:- use research-based knowledge and research methods including design of experiments, analysis, and interpretation of data and synthesis of the information to provide valid conclusions.
- PO 5. Modern tool usage:- create select and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- PO 6. The Engineer and Society:- apply to reason informed by the contextual knowledge to assossocietal, health, safety, legal, and cultural issues and consequent responsibilities relevant to the professional engineering practice.
- PO 7. Environment and sustainability:- understand the impact of the professional engineering solution in societal and environmental contents, and demonstrate the knowledge of and need for sustainable development.
- PO 8. Ethics:- Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO 9. Individual and team work:- factions effectively as an individual and as a member or leader in diverse teams and multidisciplinary settings.
- PO 10. Communications:- communicate effectively on complex engineering activities with the engineering community and with society at large such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive if clear instructions.
- PO 11. Project management and finance:- demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a memberand leader in a team, to manage projects and in multidisciplinary environments.
- PO 12. Lifelong learning:- recognize the need for and have the preparation and ability to engage in dependant and lifelong learning in the broadest context of technological changes.

A NOTE FROM THE DIRECTOR

Dear Reader, It is a great pleasure and honor for me to serve as the Director of Shri Ramswaroop Memorial College of Engineering and Management, Lucknow. It gives me sheer pleasure to know that the Computer Science & Engineering Department is going to release Departmental Magazine for Odd Semester 2023-2024.
The department has grown over the years in different aspects. The prime aim of the department is to produce skilled engineers as per the vision & mission of the Department and Institute.

The department has been NBA Accredited and has helped Institute to reach an inspirational level. I hope these initiatives will help students to learn more and more and indeed, they will compete at national and international levels in large numbers.

I appreciate the efforts of *Dr. Pankaj Kumae*, Head of Department Computer Science and Engineering, for his commitment, leadership, and visionary approach. I expect in the future he will continue in the same league.
Needless to say, the contribution of faculties in terms of student studies is praiseworthy.

Further, I would like to appreciate Editors Er. Priyanka and Er. Amit Kumar Jaiswal for presenting the magazine of the department in its present form.

Thanks and Regards,
Dr. Bhavesh Kumar Chauhan
Director,
Shri Ramswaroop Memorial College of Engineering and Management, Lucknow

MESSAGE FROM THE HEAD OF DEPARTMENT

Greetings from the Department of Computer Science & Engineering (CSE), Shri Ramswaroop Memorial College of Engineering and Management, Lucknow. It gives me immense pride and pleasure to lead the Department of Computer Science and Engineering. The CSE department primarily provides undergraduate students with knowledge and skills in the field of computer science. The department aims to provide an atmosphere that allows students to acquire analytical and practical skills and apply them to real-life problems

. The department provides an outstanding academic atmosphere with a staff of highly trained faculty members who motivate students to achieve their full prospective. CSE students are encouraged to be the best software professionals or entrepreneurs through modern and innovative teaching-learning processes.

The department strives to impart knowledge and hands-on training of the highest standard ranging from fundamentals of Computer Science, programming, core courses, and emerging areas like Artificial Intelligence, Machine Learning, the Internet of Things, Data Analytics, Cyber Security, Open-Source Technologies, etc. For the overall development of students, the department of CSE is associated with memberships of professional bodies such as IEEE, CSI, ACM, GFG, and D'Coder Club coding platforms.

Since the industry keeps evolving with new technologies, we ensure to keep up with the outside corporate world and impart knowledge amongst our beloved future technocrats!

I extend my warm wishes to all promising engineers of Computer Science & Engineering.

Dr. Pankaj Kumar
Professor & Head of Department,
Department of Computer Science & Engineering(NBA Accredited Department),
Shri Ramswaroop Memorial College of Engineering and Management, Lucknow

FACULTY DEVELOPMENT PROGRAM

Department of CSE of Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM), Lucknow has Successfully Concluded Five Days Faculty Development Program on “Recent Trends in Artificial Intelligence and Cyber Security” from 01st to 05th August 2023. The FDP provided a platform for participants to delve into various topics and acquire expertise in the rapidly evolving fields of Artificial Intelligence and Cyber Security. Faculties from outside like Babu Banarasi Das Colleges Lucknow, Teerthanker Mahaveer University Moradabad, IMRT, Lucknow, etc. participated in the FDP.

Throughout the FDP, all participants actively shared their experiences and knowledge, availing themselves of the opportunity to learn from one another. The College looks forward to continued support in upcoming development programs and encourages its faculty to stay connected with the latest advancements and trends in technology and development.

THE INNOVATION DAY

DIGITAL POSTER PRESENTATION

Organized by the *Institute Innovation Council* and the *Department of Computer Science and Engineering,* SRMCEM, this event was a remarkable blend of learning, creativity, and groundbreaking ideas.

The Innovation Day wasn't just a platform for competition—it was a celebration of imagination, collaboration, and the entrepreneurial spirit. Participants showcased digital posters that reflected not only technical skills but also innovative thinking and impactful problem-solving.

A special note of gratitude to *Dr. Pankaj Kumar,* Head of the Computer Science Department, whose inspiring words highlighted the transformative role of innovation and entrepreneurship in today's world. His insights ignited a spark in all of us to think beyond limits and dream bigger.

This enriching experience reminded us of the power of curiosity and collaboration. I'm truly thankful for the opportunity and look forward to many more moments where we can learn, innovate, and grow together. Let's continue to push boundaries, think boldly, and turn visionary ideas into reality!

GOOGLE DEVELOPER STUDENT CLUB

During the session, Usha provided a clear and detailed overview of KerasNLP, explaining how it simplifies building and training powerful NLP models. She walked the audience through key concepts such as tokenization, embeddings, and model architectures used in NLP. The session also included practical demonstrations, where attendees had the opportunity to learn how to apply KerasNLP for tasks like text classification, sentiment analysis, and more.

The event was interactive, with participants actively engaging in Q&A sessions and di
scussing how NLP can be used in real-world applications, from chatbots to recommendation systems. Usha's expert insights and practical tips inspired many students to explore the vast potential of AI and NLP technologies.Overall, the webinar was a massive success, giving attendees a strong foundation to get started with AI and NLP, while also providing them with the tools to dive deeper into the field. The event highlighted GDG on Campus' commitment to fostering learning, innovation, and skill development in emerging technologies.On January 6, 2024, GDG on Campus hosted an enriching webinar titled Getting Started with AI: Introduction to KerasNLP , led by the renowned AI expert Usha Rengaraju. The webinar was aimed at introducing participants to the world of Artificial Intelligence and Natural Language Processing (NLP) using the KerasNLP library.

FLUTTER WORKSHOP

The college-level Flutter Workshop organized by GDG on Campus on 12th and 13th March 2023 was a remarkable success, drawing over 200 students eager to explore app development. Over two days, participants engaged in interactive, hands-on sessions that introduced them to the fundamentals of Flutter, enabling them to create their first applications. Expert mentors guided the sessions, offering personalized support and resolving doubts during lively Q&A interactions. The event's success was amplified through strong engagement on social media, with posts receiving significant attention and shares. Participants received certificates of completion, boosting their confidence and professional profiles. Feedback highlighted the workshop's practical focus, inspiring many attendees to delve deeper into app development. This initiative reinforced GDG on Campus's role in promoting technical learning and creating a platform for students to excel in emerging technologies.

ABHIVYAKTI

Abhivyakti 2024, the annual cultural fest, was a vibrant celebration of art and culture. The event showcased a diverse array of talent with captivating dance performances, melodious music concerts, enthralling theatrical productions, and engaging art exhibitions. Abhivyakti 2024 not only provided a platform for students to showcase their artistic talents but also fostered a deep appreciation for the rich cultural heritage of our nation.

Gantavya 2024, the annual technical fest, was a resounding success. The event witnessed enthusiastic participation with coding marathons, robotics challenges, and design contests captivating student minds. Workshops and guest lectures by industry experts provided invaluable insights, inspiring innovation. Gantavya 2024 proved to be a platform for showcasing student talent and fostering a vibrant community of tech enthusiasts.

GANTAVYA

AKTU ZONAL FEST

The inaugural Sports Fest was successfully tested the limits of athletes across the state. Events like the grueling endurance trail run, demanding strength challenges, and collaborative teamwork events showcased exceptional athleticism and resilience. The Games fostered a spirit of sportsmanship and inspired students to push their boundaries.

FRESHER'S 2023

A heartfelt welcome to all the freshers stepping into Shri Ramswaroop Memorial College of Engineering and Management

You are about to embark on a remarkable journey of learning, discovery, and transformation. At SRMCEM, you'll become part of a dynamic community that champions academic excellence, innovation, and holistic development.

With a dedicated faculty, cutting-edge infrastructure, and an encouraging atmosphere, the college offers countless opportunities to explore your passions, hone your skills, and forge lifelong friendships. From classrooms and laboratories to cultural festivals and technical events, every experience here is designed to inspire and empower you.

As you begin this exciting new chapter, embrace it with curiosity, courage, and an open mind. Your aspirations have found the perfect place to grow — welcome aboard, and best wishes for an enriching and successful journey ahead!

MIND BYTES

STUDENT'S WRITINGS

Shubham Maurya is a B.Tech student in Computer Science and Engineering (CSE) with a specialization in Artificial Intelligence (AI) and Machine Learning (ML). He is currently pursuing his studies at Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM), Lucknow, where he also holds the distinguished position of convenor.

In this capacity, Shubham has demonstrated strong leadership and organizational skills by spearheading numerous cultural events at the college, contributing actively to the vibrant campus life. His ability to coordinate large-scale functions and manage diverse teams reflects a rare blend of strategic planning, teamwork, and communication skills that are invaluable both within and beyond the academic arena.

As a convenor, Shubham Maurya has been instrumental in orchestrating various cultural activities, which not only foster community spirit but also provide a platform for students to showcase their talents beyond academics. His involvement in these events highlights his multifaceted personality, balancing rigorous technical education with cultural engagement. Through these initiatives, he has cultivated strong interpersonal networks and has inspired fellow students to participate actively in extracurricular spheres, enriching the overall student experience.

Academically, Shubham's specialization in AI and ML positions him at the forefront of contemporary technological advancements, equipping him with skills that are highly sought after in the current digital and data-driven era. His focus on these cutting-edge fields reflects a commitment to innovation and a vision to contribute to the evolving landscape of technology. He has engaged in practical projects and collaborative research work, strengthening his expertise in areas such as data analytics, predictive modeling, and intelligent systems.

In addition to his academic pursuits and cultural leadership, Shubham has shown a keen interest in community service and student welfare activities, demonstrating a commitment to holistic development. His participation in various technical workshops, coding competitions, and seminars further underscores his dedication to continuous learning and staying abreast of emerging trends in the AI-ML ecosystem.

Shubham Maurya

As a student currently pursuing a Bachelor of Technology in Computer Science and Engineering, with a specialization in Artificial Intelligence and Machine Learning (AIML) and an Honors degree in Cyber Security, I've had the opportunity to explore some of the most dynamic and impactful areas in the world of technology. From the very beginning of my academic journey,

I have been deeply fascinated by the question: Can machines think, learn, and make decisions as we do? This curiosity has driven my growing interest in the field of artificial intelligence, particularly in how intelligent systems can be designed to solve real-world problems. My core academic interests lie in machine learning, intelligent systems, and natural language processing (NLP)—domains that have the power to reshape industries, improve human lives, and push the boundaries of what machines can achieve. As I've progressed through my coursework and projects, I've found myself increasingly drawn toward the research aspect of AI—not just using existing models, but understanding their inner workings, limitations, and potential for advancement. I aspire to contribute to research that builds more robust, ethical, and efficient AI systems capable of making a real-world impact.

Beyond the classroom, I've actively engaged in coding competitions and hackathons, which have sharpened my analytical thinking and problem-solving capabilities. In the global arena, I secured Rank 2670 in Round 1 and Rank 1778 in Round 2 of TCS CodeVita Season 11, a highly competitive programming contest with participation from thousands of students across the world. These experiences helped me refine my logical thinking, strengthen my grasp of algorithms, and build solutions under tight deadlines—skills essential for any aspiring AI researcher or developer.

Leadership and community involvement have also played a crucial role in shaping my college life. During my first year, I served as a student council member, which provided valuable exposure to team coordination, event planning, and student engagement. These responsibilities taught me the importance of initiative, clear communication, and leading by example.

Striking a balance between academic rigor, extracurricular involvement, and skill development has been both challenging and rewarding. Each initiative I take—whether it's a coding contest, a personal project, or a research paper—adds to a larger goal of building a well-rounded, purpose-driven, and intellectually curious version of myself.

Looking ahead, I am deeply motivated to pursue research in AI, particularly in areas that explore the interaction between learning algorithms and real-world data. I envision myself contributing to projects that not only push technical boundaries but also address ethical and societal concerns around AI deployment. My goal is to become not just a skilled engineer, but a responsible innovator—someone who can contribute meaningfully to the future of intelligent technology.

Isha Ojha

In the dynamic academic environment of Shri Ramswaroop Memorial College of Engineering and Management (SRMCEM), Himanshu Verma emerges as a distinguished figure whose academic rigor, technical expertise, and leadership acumen set a benchmark for aspiring engineers. A third-year student of Computer Science and Engineering (Artificial Intelligence & Machine Learning),

Himanshu has continually demonstrated a commitment to harnessing technology for meaningful applications. His academic pursuits are complemented by an impressive array of practical projects and professional trainings that exemplify his proactive approach to learning.

Himanshu undertook comprehensive training on the MERN stack (MongoDB, Express.js, React.js, Node.js) from SoftPro India Pvt. Ltd., during which he successfully developed MUSCLEMETER, a health-oriented application designed to promote fitness through modern web technologies. His flair for applied learning is further showcased by his FOOD project, an initiative completed during an online internship with Tutedude, focusing on promoting healthy dietary habits through intuitive digital solutions.

Currently, Himanshu is expanding his expertise into the field of Data Science, deepening his knowledge in AI-driven data analysis and predictive modeling, further enriching his profile as a future-ready engineer.

In addition to his technical accomplishments, Himanshu has actively contributed to the organisational landscape of SRMCEM. Serving as both Coordinator and Head Coordinator in various college committees, he has exhibited exemplary leadership, teamwork, and event management capabilities — attributes that reflect his well-rounded personality and dedication to community-building.

A charismatic individual with a charming personality and a relentless pursuit of excellence, Himanshu Verma embodies the qualities of a modern technocrat poised to make meaningful contributions in the evolving landscape of artificial intelligence and machine learning.

Adept at coding, driven by efficiency in learning, and passionate about innovation, Himanshu Verma stands as a beacon of inspiration to his peers. As he continues his journey, his story is one of perseverance, purpose, and the pursuit of meaningful impact — a testament to the evolving landscape of AI and the limitless potential of those who dare to shape it.

Himanshu Verma

I'm Keshav Bajpai, a third-year Computer Science student with a deep interest in the history and culture of India. While my academic pursuits revolve around technology, I've always been fascinated by the profound legacy of Indian civilization. The country's rich history, from the ancient Indus Valley civilization to the diverse empires that followed, showcases a unique blend of art, science, philosophy, and spirituality that continues to influence the world today. In this article, I aim to delve into the many facets of Indian civilization, exploring its contributions to fields like mathematics, astronomy, and literature, as well as its rich cultural and spiritual heritage. By reflecting on its ancient wisdom, I hope to provide insight into how India's historical achievements remain relevant in the modern era and offer valuable lessons for the present and future.

India: A Civilization Remembered

Civilizations rarely collapse—they drift, slowly forgetting the fire that made them. India once shaped time with zero, mapped the soul with Upanishads, and questioned the cosmos before telescopes existed. But waves of conquest, colonization, and cultural erosion dimmed that brilliance. The British didn't just seize land—they rewrote identity. For decades, we wore borrowed languages and doubted native thought.

But memory is stubborn.

In the 21st century, India isn't reviving old glory—it's remixing it. Code now echoes ancient logic. AI systems draw from dharma. Startups rise from cities that once housed sages. We are no longer a civilization playing catch-up—we're scripting our own future, rooted in something deeper.

This isn't nostalgia. It's a return to coherence. A rediscovery of who we were, so we can become who we're meant to be.

Keshav Bajpai

Coordinators Team

Faculty Coordinators

Dr. Pankaj Kumar

Er. Himanshi Singh

Editorial and Design Team

ISHA OJHA
(3rd Year)

SHUBHAM MAURYA
(3rd Year)

RISHU MISHRA
(3rd Year)

HIMANSHU VERMA
(3rd Year)

UNNATI GUPTA
(2nd Year)

KRISHNA KUSHWAHA
(2nd Year)